

CSC490

Assignment 1

Project Name: VocaLocate

Group Members:

Riyad Valiyev (Student Number: 1009069874)
Boquan Zhao (Student Number: 1010419343)
Tanayjyot Singh Chawla (Student Number: 1010411109)
Carmen Chau (Student Number: 1009043376)

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Part One: Interest Statements

Sound effects (SFX) are important media storytelling components, but many editors struggle to find the right sound in their growing SFX libraries. Folder navigation is slow, while filename/metadata based keyword search may not capture acoustic qualities editors have in mind (Zhang & Duan, 2016). We aim to address this problem with VocaLocate, which uses vocal imitations and optional semantic text descriptors to retrieve closest SFX matches from a user's local collection. We are passionate about improving creative workflows and excited to learn how multimodal ML can make SFX retrieval more intuitive and efficient. Here are our interest statements:

Carmen: I am fascinated by the power sound effects hold in shaping emotional responses and enriching user experiences. This motivates me to explore what acoustic characteristics people retain and reproduce when recalling impactful sound effects from memory. This is why I want to explore audio preprocessing and audio encoder fine-tuning for similarity-based SFX retrieval.

Boquan: I am interested in implementing the voice-and-text hybrid retrieval approach and then tuning and evaluating it to see how well the two models work together, that process is something I have not had much experience with in my previous ML development work.

Riyad: What draws me in is the engineering challenge of making local indexing and retrieval infrastructure feel instant and trustworthy on a user's own private library.

Tanay: I would love to build evaluations for our model and compare it against existing benchmarks. I am especially interested in seeing whether the improvements we make actually lead to better retrieval.

Part Two: Landscape Analysis

Relevant Item	Description	Commentary (citations for all rows are included in the Appendix)
Soundly [Link]	Commercial SFX management platform. Can search SFX from personal collection + Soundly's catalog via text descriptors and reference audio, with direct editor integration.	Soundly does not document specialized SFX retrieval using vocal imitation. VocaLocate can differentiate by using vocal-imitation retrieval, adopting Soundly's domain-specific text search, and emphasizing minimal workflow friction.
Sononym [Link]	Commercial audio sample manager that finds similar sounds from user libraries via keyword/tag search over filenames and metadata + similarity search via existing audio or live recordings.	Sononym shows that local audio similarity search is feasible, but it relies on general acoustic similarity and existing textual labels. This motivates VocaLocate to adopt a local-first approach while specializing retrieval based on vocal imitation and semantic text refinement.
Storyblocks (specifically: "Search by Sound" feature) [Link]	Commercial stock-media platform that can find music tracks and SFX from uploaded reference audio clips and metadata keyword search.	Storyblocks shows reference-audio similarity search scales well to large catalogs but struggles to robustly process imitation-style queries. This highlights VocaLocate's need to optimize the preprocessing of noisy vocal queries.

UnstuntedSFX [Link]	Commercial SFX library that markets a "hum-to-search" (eg: hum, whistle, clap) search over its own catalog. Includes editor tool plugin integration.	Close functional similarity to our project, but does not support search over a user's SFX collection. The website is not fully functional, reinforcing the importance for us to build reliable SFX retrieval over a private library.
thatsoundslikeme [Link]	A 2025 research project with web and desktop application where users record vocal imitations and retrieve SFX matches from a public Freesound dataset.	thatsoundslikeme lacks local-library and hybrid voice+text search, but its MobileNetV3 embeddings, cosine-similarity retrieval, and lightweight deployment outline a useful technical precedent for VocaLocate.
Voogle [Link]	A 2019 research prototype (local web app). Users upload vocal imitations or audio clips to search files in any audio collection that is uploaded to the platform.	Voogle lacks hybrid voice+text search and library re-indexing. Its separate vocal-imitation and SFX encoders are a useful benchmark to understand Querying Sound Effects by Vocal Imitation (QVIM) fundamentals.
"Improving Query-by-Vocal Imitation with Contrastive Learning and Audio Pretraining" [Link]	Research paper exploring the development and fine-tuning of pretrained MobileNetV3 encoders to pre-process vocal imitations and reference SFXs.	Demonstrates that pretrained audio encoders with contrastive fine-tuning substantially improve vocal-imitation retrieval, providing a foundation for VocaLocate's MobileNetV3 model and further fine-tuning experiments.
"CLAP: Learning Audio Concepts From Natural Language Supervision" [Link]	Research paper introducing CLAP, an approach for mapping text-to-audio matching in a shared embedding space for semantic similarity sound search.	CLAP illustrates that natural-language descriptors can be descriptively aligned with audio without reliance on predefined labels, and performs well across diverse audio domains. Thus, it's important for us to explore its use for semantic text-to-audio matching in VocaLocate.
"Multimodal Fusion for Multimedia Analysis" [Link]	Survey of methods for combining information from multiple modalities, including decision-level linear weighted fusion.	Identifies normalized weighted fusion as a simple, widely used way to combine independent modality scores. This supports VocaLocate's plan to normalize QVIM and CLAP similarity scores and tune their weights for the final ranking.
VimSketch Dataset [Link]	Public QVIM Dataset with 542 reference sounds and 12,543 vocal imitations. 13–37 imitations per reference.	Provides paired imitation-reference data across diverse sounds relevant to SFX search. This dataset is strong starting point for training + fine-tuning VocaLocate's vocal-retrieval model.

Part Three: Project Outline

Problem Statement: Sound libraries for film, game, and podcast production often contain hundreds of thousands of files with uninformative names (e.g., SFX_0231.wav). Editors usually know what a sound should sound like better than they can name it, so they page through keyword results or scrub through folders by ear, which is slow and error-prone. Existing tools cover nearby cases. Soundly's "Find Similar" searches using a clip the editor already has, and UnstuntedSFX's hum/whistle search lets editors search its own catalog by imitation. Neither helps an editor who has no reference clip and needs to search their own private or purchased library, which usually can't be uploaded to a third party for licensing and confidentiality reasons.

Proposed Solution: VocaLocate lets an editor imitate a sound with their voice, optionally add a short text qualifier ("metallic," "short"), and get a ranked list of the closest matches from their own SFX library. Each file is indexed once on the user's machine, and new files are added as they arrive, so the library is never uploaded or reprocessed from scratch. When the user searches, a vocal-imitation model compares their voice to each sound, a text-audio model compares any qualifier to the same sounds, and the two scores are combined into a single ranking. The index also records which project or folder each sound came from, so a user can recover a sound they remember using in an earlier project. Where Voogles is an outdated research prototype, thatsoundslikeme searches a fixed public dataset, UnstuntedSFX searches only its own catalog, and Soundly cannot process vocal imitation, VocaLocate brings modern imitation search to the user's own library.

High-Level Technical Approach and Milestones: We use two MobileNetV3 audio encoders pretrained on AudioSet, a large general-purpose sound dataset. One encoder handles vocal imitations and the other handles real sound effects. We fine-tune them with contrastive learning on VimSketch, a public dataset of imitations paired with the sounds they imitate, so that an imitation and its matching sound end up with similar embeddings (numerical representations of each sound). At search time, cosine similarity between the query's embedding and each file's embedding produces a voice score. Optional text is handled by a pretrained CLAP model, used without fine-tuning, which gives each sound a text score from the cosine similarity between the text and the sound's audio embedding. After both scores are normalized to the same scale, they are merged with a tuned weight $s = \alpha \cdot s_{\text{voice}} + (1 - \alpha) \cdot s_{\text{text}}$. Our main training experiment is hard-negative mining: deliberately showing the model the incorrect sounds it most easily confuses with the correct one, to test whether this teaches it finer distinctions. Embeddings are stored in a local vector index (FAISS), which keeps search fast and lets new files be added without rebuilding the index. Our milestones: M1 reproduces the pretrained MobileNetV3 baseline and a simple non-neural baseline. M2 sets up data splits that keep each reference sound in a single split and runs fine-tuning experiments. M3 adds hard-negative mining and CLAP text scoring. M4 builds the local indexing and search app with project tagging. M5 runs a comparison study with real editors and finalizes results.

Unknowns to Investigate:

- Whether real editors find sounds faster with vocal imitation and project history than with text or reference-audio search.
- Whether hard-negative mining improves retrieval at our data and compute scale.
- Whether CLAP text scores improve results, given CLAP was not trained alongside our audio model.
- Whether rank-based or learned fusion outperforms a simple weighted sum.
- How search speed holds up as a library grows past 100,000 files.
- Whether one shared model works across users' different imitation styles.

Part Four: Project Press Release

VocaLocate: Find the Sound in Your Head, in the Library You Already Own

For video editors, game audio designers, and podcast producers who know exactly what sound they are looking for but can't find it in their own sound effect library.

TORONTO, ON - December 7, 2026 : The VocaLocate team released the free public beta of VocaLocate, a desktop app that lets editors search their sound effect (SFX) library by recording vocal imitations (eg: "whoosh," a "clunk," or "eeeeek"). If you like, add some descriptive words such as "metallic" or "short." VocaLocate then shows the closest-sounding files from the audio collection on your computer.

Over time, editors and sound designers build large SFX libraries from purchased packs and past projects. When they want to retrieve an SFX to use, they usually have a sound in mind but won't remember the exact file name or location. So they try searching using keywords or listening through files one by one. This can lead them to settle for a worse match or even repurchase a sound they already own.

With VocaLocate, you choose the sound folders you want to search. The app organizes them in the background and keeps the searchable library up to date as you add new files. To search, record an imitation with your voice and let VocaLocate find the closest matching sound effect files, ranked by how close they sound to your imitation and your text description. VocaLocate shows where each matched file is stored on the computer, so users can quickly reuse existing audio in their editing workflows.

Today, editors usually search by keywords in tools like Soundly or by using an existing reference clip in tools like Storyblocks. Keyword search depends on useful filenames and tags, while reference-audio search only works when the user has a similar clip. Vocal-imitation search exists, but it's rarely seen in many publicly available tools and is still concentrated in research prototypes, which typically search fixed public sound collections rather than users' own libraries. VocaLocate fills this gap by bringing hybrid voice-and-text search to help users search within their local sound effect collection, keeping proprietary sound packs and personal recordings on the user's machine.

"Every editor has had a sound stuck in their head and no way to type it into a search box," said Riyad, co-creator of VocaLocate. "We built VocaLocate so you can just make the sound and get back to editing."

"I have eight years of sounds spread across old projects, and half of them are named things like 'final_v3,'" said Maya, an independent game audio designer. "I used to lose 20 minutes hunting for one sound. This time I made a rattling noise into my mic and found a chain sound I recorded three years ago in seconds that I didn't remember owning."

"Our contracts don't let us upload our licensed libraries to outside services," said Daniel Reys, post-production lead at a small Toronto studio. "VocaLocate searching everything locally is what made it possible for us to try it at all."

Get started today! Download VocaLocate at VocaLocate.app, choose the folders you want to search, and let it index them in the background. You can start searching by voice right away while it finishes.

Appendix - Press Release Iterations

Iteration 2:

VocaLocate: Find the Sound in Your Head, in the Library You Already Own

For video editors, game audio designers, and podcast producers who know exactly what sound they are looking for but can't find it in their own sound effect library.

TORONTO, ON December 7, 2026: Today the VocaLocate team released the free public beta of VocaLocate, a desktop app that lets editors search their own sound effect library by imitating a sound out loud. Make a "whoosh," a "clunk," or a creaky-door "eeeeek" into your mic. If you like, add a word or two such as "metallic" or "short." VocaLocate then shows the closest-sounding files from your own collection. Your sounds never leave your computer.

Over the years, editors and sound designers build up huge sound libraries: packs they bought, recordings from old projects, and random downloads. That can add up to tens of thousands of files with names like SFX_0231.wav. When they need a sound, they usually know exactly what it should sound like but not what it's called. So they type keywords that don't really describe it, or click through folders listening to one file at a time. This can eat up a big part of an editing session. Often they give up and settle for a worse sound, or buy a sound they already own.

With VocaLocate, you point the app at your sound folders once. It works through every file in the background and keeps up as you add new ones. To search, hold a key, make the sound with your voice, and let go. VocaLocate ranks your files by how close they sound to your imitation. You can add a text hint to narrow things down, or search only one project's folder. VocaLocate also remembers which project each sound came from, so you can find "that impact from last spring's trailer" again. You can preview any result in the app and drag it straight into your editing software.

Today, editors search with keywords in library tools like Soundly or BaseHead. Some use "search by sound" features like Splice's Search with Sound, which only work if you already have a similar clip. A few services let you hum to search, but only inside their own catalogs. None of them let you search, by voice, the library you already own and paid for. VocaLocate does, and it runs entirely on your machine, so licensed sound packs and private studio recordings stay private.

"Every editor has had a sound stuck in their head and no way to type it into a search box," said the VocaLocate team. "We built VocaLocate so you can just make the sound and get back to editing."

"I have eight years of sounds spread across old projects, and half of them are named things like 'final_v3,'" said Maya Okafor, an independent game audio designer. "I made a rattling noise into my mic and found a chain sound I recorded three years ago. I had completely forgotten I owned it."

"Our contracts don't let us upload our licensed libraries to outside services," said Daniel Reyes, post-production lead at a small Toronto studio. "VocaLocate searching everything locally is what made it possible for us to try it at all."

Getting started takes a few minutes. Download VocaLocate at [VocaLocate.app](#), choose the folders you want to search, and let it index them in the background. You can start searching by voice right away while the model finishes indexing. VocaLocate is free during the beta.

Iteration 1:

VocaLocate: Find the Sound in Your Head

For editors who know exactly how a sound should sound but struggle to find it in their own sound effect library.

TORONTO, ON December 7, 2026 Today the VocaLocate team released the free public beta of VocaLocate, a desktop app that lets editors search a sound effect library by imitating a sound out loud. Make a "whoosh," a "clunk," or a creaky-door "eeeeek" into your mic. VocaLocate then shows the closest-sounding files from the library.

Over the years, editors and sound designers build up huge sound libraries: packs they bought, recordings from old projects, and random downloads. That can add up to tens of thousands of files with names like SFX_0231.wav. When they need a sound, they usually know exactly what it should sound like but not what it's called. So they type keywords that don't really describe it, or click through folders listening to one file at a time. This can eat up a big part of an editing session. Often they give up and settle for a worse sound, or buy a sound they already own.

With VocaLocate, users point the app at a collection of sound files. To search, hold a key, make the sound with your voice, and let go. VocaLocate ranks the files by how close they sound to the imitation. Users can preview the results directly in the app and choose the sound that best matches what they had in mind.

Today, editors mainly rely on keywords and manually previewing files when searching for sound effects. VocaLocate provides another way to search when the user knows what they want to hear but does not know what to call it.

"Every editor has had a sound stuck in their head and no way to type it into a search box," said the VocaLocate team. "We built VocaLocate so you can just make the sound and find it."

"I have eight years of sounds spread across old projects, and half of them are named things like 'final_v3,'" said Maya Okafor, an independent game audio designer. "Being able to imitate the sound I want would make searching through those files much easier."

Getting started takes a few minutes. Download VocaLocate at [VocaLocate.app](#), choose the folders you want to search, and start searching by voice.

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